

KOKOMO MUNICIPAL, IN, USA

WMO: 724387

Lat: 40.528N

Lon: 86.059W

Elev: 830

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 54807

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.6	(c) 6.9	(d) -8.3	(e) 3.6	(f) 2.4	(g) -2.5	(h) 5.0	(i) 8.2	(j) 32.0	(k) 27.8	(l) 28.7	(m) 29.0	(n) 9.2	(o) 280	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	90.9	74.7	88.8	73.8	86.3	72.5	77.5	87.6	76.1	85.1	74.6	83.0	11.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	133.1	83.5	72.9	125.8	81.5	72.0	122.0	80.4	41.3	87.4	40.1	85.9	38.5	82.4	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	23.7	20.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	-3.6	79.8	11.3	2.3	-11.7	81.4	-18.4	82.7	-24.7	84.0	-33.0	85.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	27.8	28.9	41.3	52.0	63.7	71.5	74.1	72.3	67.0	54.9	42.5	32.5
	DBStd	18.96	12.08	11.82	11.05	8.88	8.63	6.03	5.42	4.94	6.83	8.94	9.79	9.95
	HDD50	2549	693	595	317	78	6	0	0	0	0	50	264	546
	HDD65	5533	1154	1011	739	397	131	15	3	5	57	336	677	1008
	CDD50	3465	4	5	46	139	431	646	748	691	511	201	39	4
	CDD65	969	0	0	3	8	91	211	286	231	117	21	1	0
	CDH74	8676	0	0	33	100	828	1786	2676	1933	1116	200	4	0
	CDH80	2841	0	0	5	8	250	599	973	588	369	49	0	0
Wind	WSAvg	9.7	12.2	11.4	11.3	11.6	10.1	8.4	6.9	6.2	7.0	9.2	10.6	11.1
Precipitation	PrecAvg	43.8	2.6	2.3	3.0	4.1	4.4	5.2	4.8	4.2	3.8	3.2	3.3	2.9
	PrecMax	60.2	7.0	5.7	5.9	10.0	7.7	14.6	14.0	7.9	8.2	7.0	6.6	6.7
	PrecMin	30.2	0.9	0.4	0.7	1.1	1.4	1.1	1.5	1.0	0.9	1.0	0.8	0.8
	PrecStd	7.2	1.3	1.4	1.2	2.0	1.6	2.6	2.6	1.7	1.7	1.7	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.6	63.0	79.4	81.5	91.1	92.9	95.0	91.1	91.3	85.6	72.7	60.9
		MCWB	56.5	54.3	63.7	62.2	70.5	73.6	77.2	76.2	73.5	71.5	59.5	55.8
	2%	DB	54.3	56.5	69.7	77.3	86.0	90.1	90.8	88.4	88.4	80.5	66.0	55.2
		MCWB	50.8	52.5	59.0	60.7	69.1	73.2	76.2	75.3	72.5	66.4	58.6	51.7
	5%	DB	49.4	50.3	63.3	73.1	82.2	87.6	88.3	85.8	84.2	74.7	62.7	50.8
		MCWB	47.0	46.8	54.6	59.0	67.1	72.1	74.7	73.7	70.3	63.7	55.5	47.5
	10%	DB	44.5	44.8	57.5	69.2	79.3	83.4	85.5	82.8	81.2	70.4	57.3	45.7
		MCWB	40.3	40.6	51.6	57.9	66.0	69.9	73.1	72.0	66.9	61.8	51.0	42.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.6	56.8	64.1	66.1	73.5	77.7	80.2	78.8	76.8	73.0	63.7	56.7
		MCDB	60.4	59.5	77.3	76.8	82.8	89.2	89.2	88.2	87.6	82.3	69.8	60.4
	2%	WB	51.0	51.9	59.9	63.5	71.4	75.7	78.1	76.9	74.4	69.4	59.3	52.6
		MCDB	53.3	56.7	67.4	73.1	81.8	85.1	88.0	85.8	84.3	76.1	64.1	55.2
	5%	WB	46.3	46.2	56.2	60.9	69.4	73.9	76.4	75.4	72.2	66.6	56.5	47.0
		MCDB	48.9	49.0	61.9	70.3	79.6	83.2	85.6	83.4	81.2	72.3	62.5	49.9
	10%	WB	40.4	40.4	52.6	58.6	67.7	72.1	74.7	73.6	69.7	62.5	51.5	42.1
		MCDB	43.8	44.1	58.1	68.1	76.7	80.8	83.0	80.6	77.2	69.2	56.8	45.5
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.7	16.9	20.8	19.8	19.2	19.6	19.5	22.0	19.5	15.6	12.6
		MCDBR	17.4	20.6	23.2	27.2	23.6	23.0	21.3	21.9	25.8	25.3	20.8	18.0
		MCWBR	16.0	17.6	16.2	16.1	12.3	10.7	10.7	12.2	12.5	14.3	15.5	16.1
	5% WB	MCDBR	16.2	19.7	20.3	22.7	20.2	18.1	18.7	18.2	21.3	20.3	18.7	17.6
		MCWBR	15.4	17.7	15.2	15.9	11.7	9.8	10.3	10.9	11.4	12.8	15.6	16.3
Clear-Sky Solar Irradiance	taub	0.300	0.327	0.347	0.409	0.450	0.437	0.451	0.451	0.415	0.358	0.321	0.301	
	taud	2.410	2.321	2.358	2.188	2.119	2.206	2.175	2.160	2.243	2.399	2.455	2.443	
	Ebn at Noon	273	280	287	274	264	267	262	258	260	266	263	263	
	Edh at Noon	26	33	35	45	49	45	46	46	39	30	25	23	
All-Sky Solar Radiation	RadAvg	534	810	1148	1490	1713	1942	1938	1730	1418	968	633	443	
	RadStd	55	83	89	138	156	129	111	97	112	89	73	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page